Sawbones Solid Rigid Polyurethane Foam 5 pcf Density

Categories: Other Engineering Material; Composite Core Material; Polymer; Thermoset; Polyurethane, TS; Thermoset Polyurethane Foam, Unreinforced

Material Notes:

Solid Rigid Polyurethane Foam

Primary Use
Solid rigid polyurethane foam is primarily used as an alternative test medium for human cancellous bone. These products are not intended to replicate the mechanical properties of human bone, however, it does provide consistent and uniform material with properties in the range of human cancellous bone. Relevant mechanical properties for comparison to human cancellous bone may depend on the particular test method that is being developed.

The ASTM F-1839-08 "Standard Specification for Rigid Polyurethane Foam for Use as a Standard Material for Testing Orthopaedic Devices and Instruments" states that, "The uniformity and consistent properties of rigid polyurethane foam make it an ideal material for comparative testing of bones screws and other medical devices and instruments."

Description
Solid rigid polyurethane foam has a closed cell content ranging from 96.0 to 99.9%. Foam is available in a range of sizes and densities, from 0.08 to 0.80 grams per cubic centimeter (5 to 50 pounds per cubic foot).
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Standard block size: 13cm X 18cm X 4cm
Standard sheet size: 13cm X 18cm X 3mm

Other Notes
Water absorption ranges from 0.301 to 0.0 kg/m^2
Foam meets ASTM F-1839-08. Block dimensions are within ±2mm and sheet thickness is within ±0.3mm.
Foam density may vary ±10%.
Properties parallel to direction of the rise of foam.

Information provided by Sawbones.

Vendors: No vendors are listed for this material. Please click here if you are a supplier and would like information on how to add your listing to this material.

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Physical Properties	Metric	English	Comments
Density	0.0800 g/cc	0.00289 lb/in³	

Mechanical Properties	Metric	English	Comments
Tensile Strength	1.00 MPa	145 psi	
Tensile Modulus	0.0320 GPa	4.64 ksi	
Compressive Yield Strength	0.600 MPa	87.0 psi	
Compressive Modulus	0.0160 GPa	2.32 ksi	
Poissons Ratio	0.30	0.30	Calculated
Shear Modulus	0.00710 GPa	1.03 ksi	
Shear Strength	0.590 MPa	85.6 psi	

Thermal Properties	Metric	English	Comments
CTE, linear	63.0 µm/m-°C @Temperature -46.0 - 93.0 °C	35.0 µin/in-°F @Temperature -50.8 - 199 °F	

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